Manufacturing Quality – Catching the Bad Chip from 590 Sensors

Focus engineering investigation on the three sensors first

MSDS696 Data Science Practicum II

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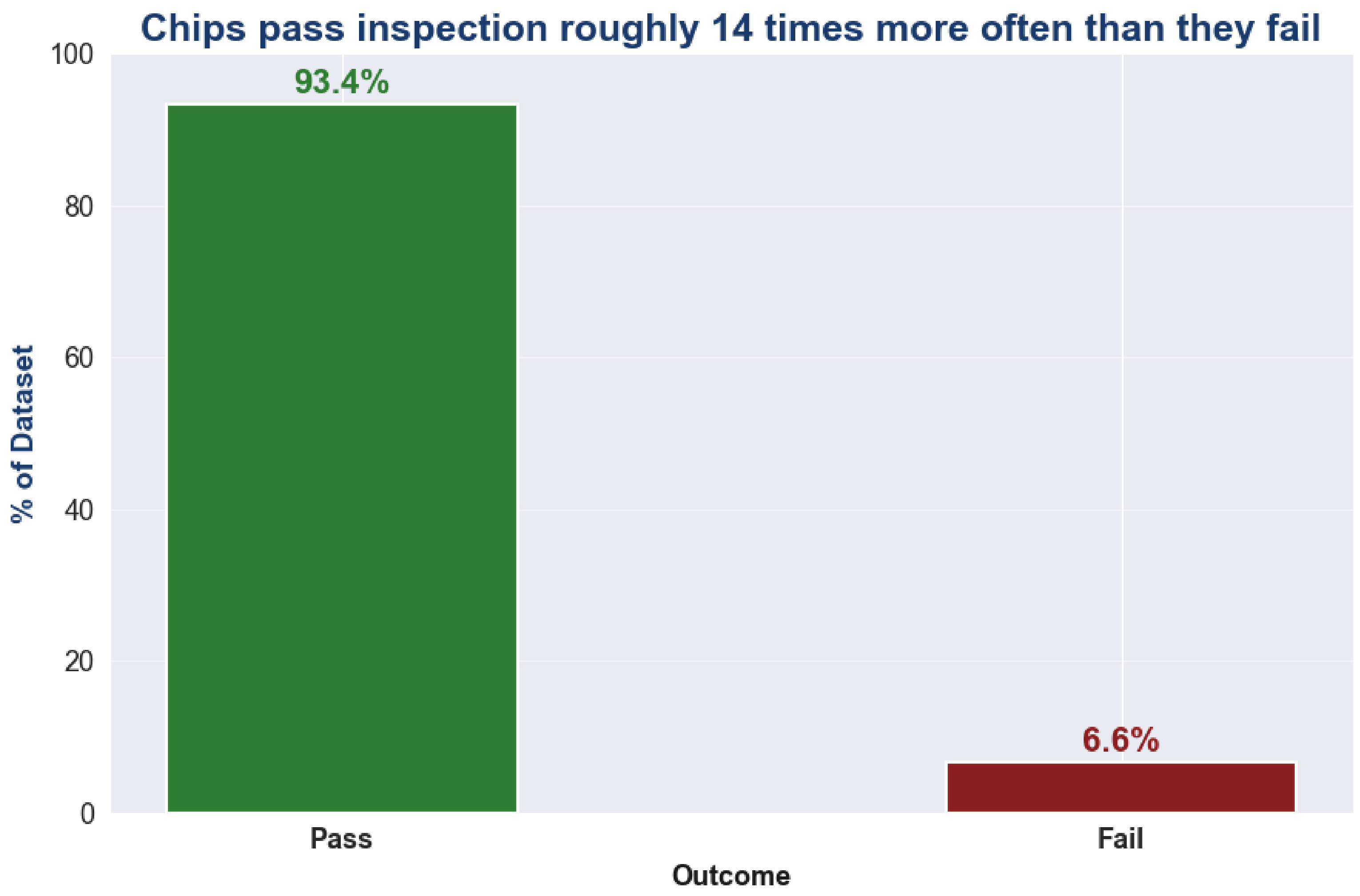
This briefing is for the executives who decide where engineering investigation hours go

Not a modeling walkthrough, a recommendation for where to spend a scarce resource, backed by evidence.

Only 6.6% of chips fail inspection, making detection difficult



Out of 1,567 chips, only 104 failed. Rare enough that guessing “Pass” every time is right 93% of the time – and catches zero defects

Most of the 590 sensors don’t actually help detect defective chips

590
Sensors collected

→

75
Statistically discriminative

→

10
Strongest warning signs

→

3
Recommended starting points

Across 1,567 chips, only 75 of 590 sensors show a meaningful, statistically confirmed difference between Pass and Fail chips. The other 515 are noise for this question

The model catches roughly 1 in 4 defective chips while flagging only about 5% of good chips for extra inspection

~1 in 4	~5%
Defective chips caught	Good chips flagged (false alarms)

Random Forest, tested once on 314 chips it never saw during training, up from just 7% recall at the default settings. Nearly 4x better than the 6.6% you would catch by chance.

An independent statistical test agrees with the model on which sensors matter

THE MODEL SAYS

feature_59

The sensor the model relies on most heavily to distinguish Pass from Fail chips apart

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THE STATISTICS SAY (INDEPENDENTLY)

feature_59

The sensor the standard statistical test choose to tell Pass from Fail chips apart

The model's ranking could just be an artifact of how it was trained. So, the top sensors were checked a second, completely different way, a standard statistical test comparing Pass and Fail chips directly, with no knowledge of the model's predictions.

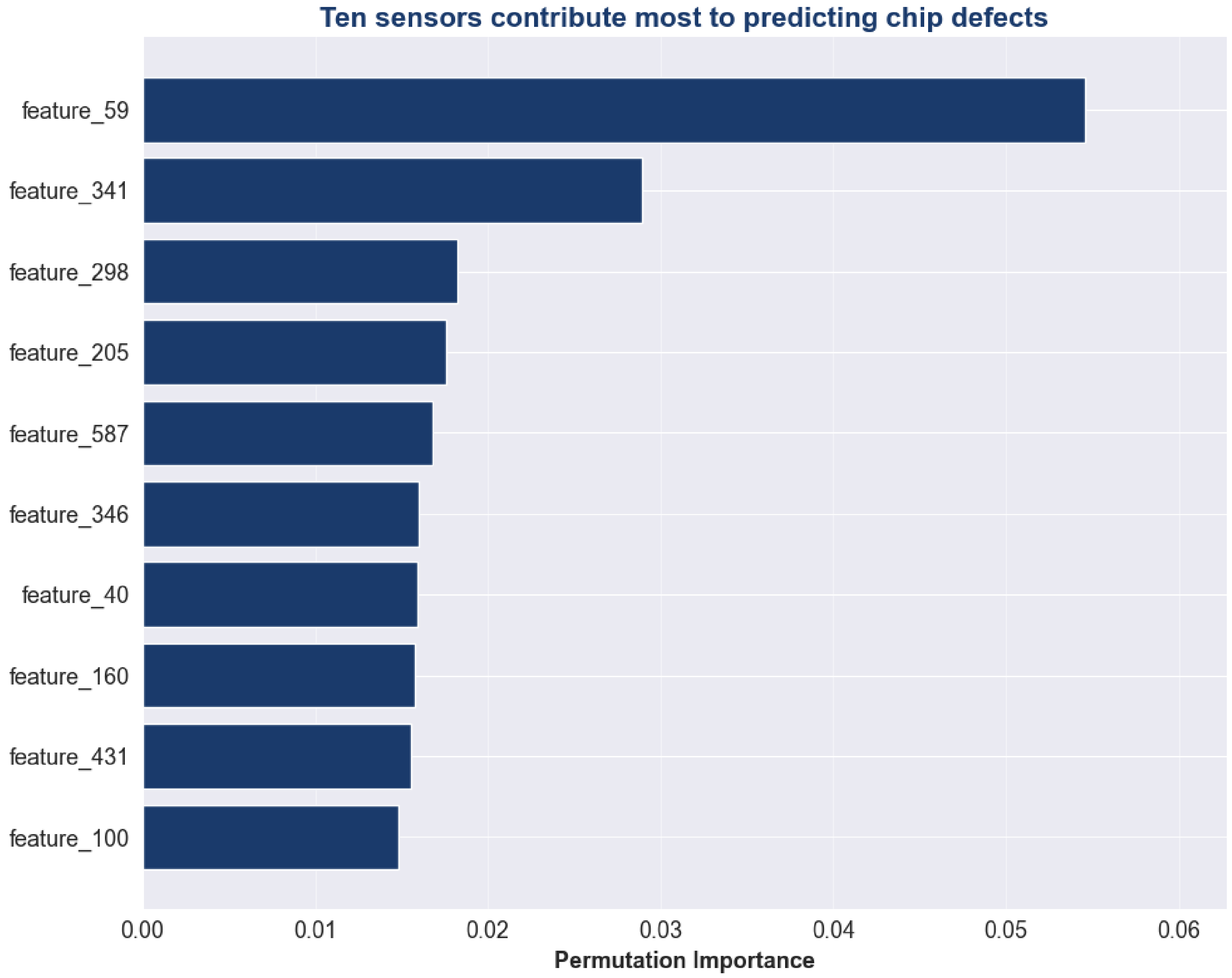
feature_59, feature_341, and feature_298 explain most of the defect signal – start there

feature_59

feature_341

feature_298

Their actual process meaning must be validated against manufacturing records.



Start with three sensors that provide the strongest warning signs of defective chips

What to do	Start with feature_59, feature_341, and feature_298 - validate them against process logs, and use the model to prioritize inspection queues, not replace them.
What it’s worth	Defects caught before they ship jump from near-zero to roughly 1 in 4 - at a false-alarm rate light enough for the inspection line to absorb.
Confidence	Independently confirmed by a second statistical test, not just the model's own ranking.

These results prioritize investigation – they do not establish root cause

Small failure sample

Only 104 failed chips total. Rankings could shift as more production data comes in.

Priority, not proven cause

This ranks where to look first, not why chips fail - validate before assuming these sensors are the root cause.

The recommendation stands: start with feature_59, feature_341, and feature_298.

Q&A

Appendix

High dimensionality and missing data make model building challenging

The goal was to improve data quality while preserving as much information as possible.

Raw Data — 590 sensors × 1,567 chips

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Remove sensors with excessive missing data

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Median imputation

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Remove low-information variables

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Select strongest predictors

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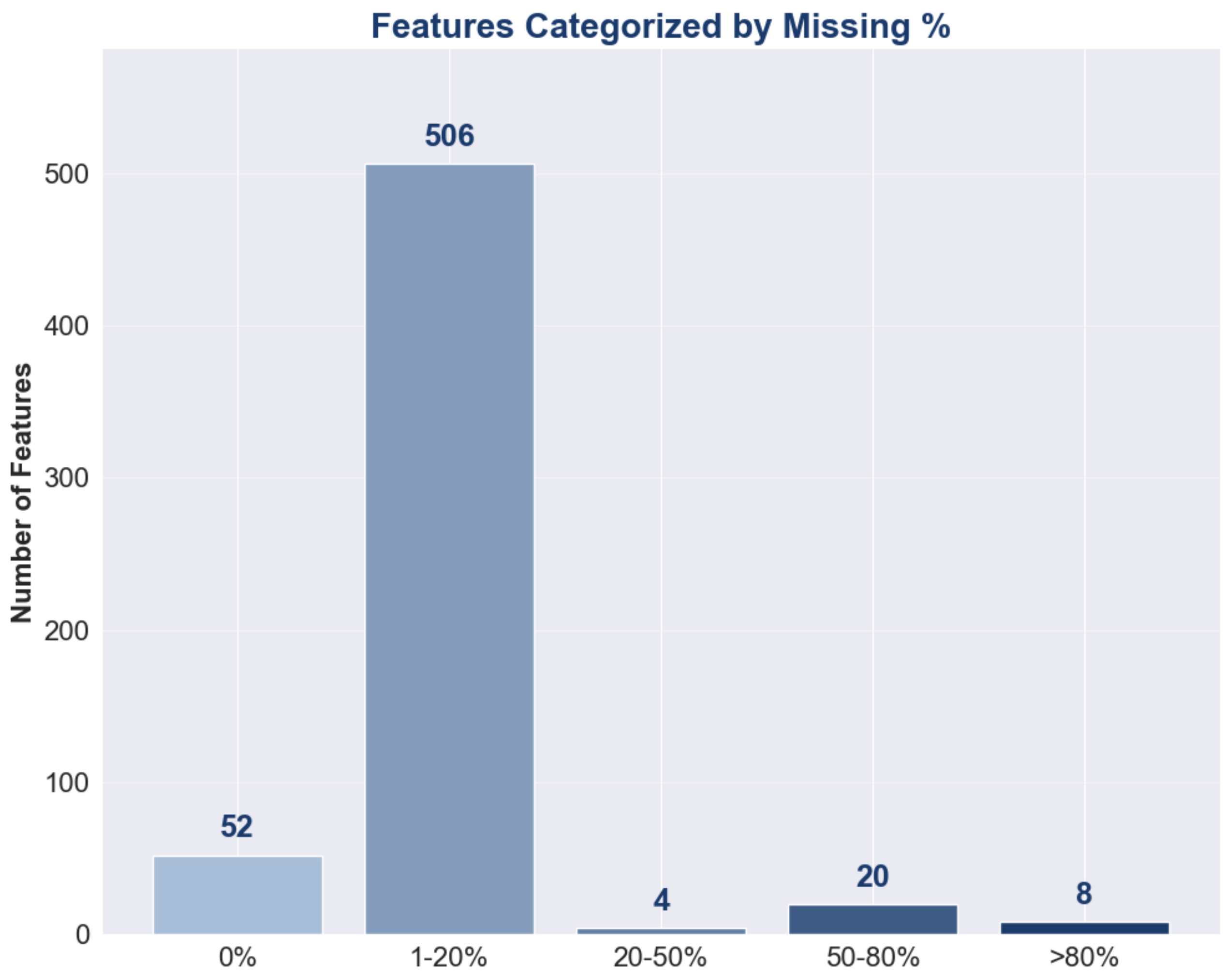
Model comparison — 99 final features



Several machine learning approaches were compared

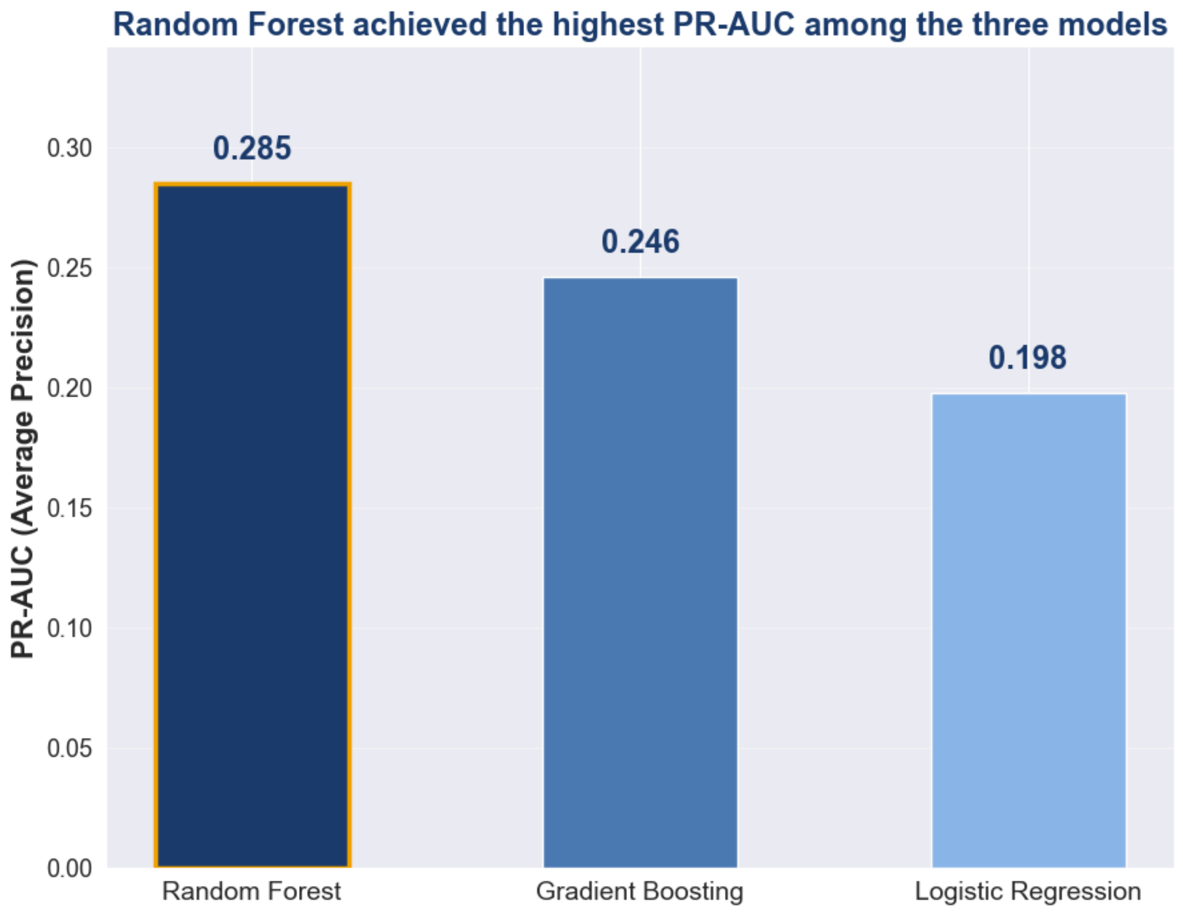
- **Logistic Regression:** A simple, interpretable baseline to confirm the signal is not linear.
- **Random Forest:** Handles non-linear sensor interactions and is robust to the noisy, high-dimensional feature set.
- **Histogram Gradient Boosting:** A stronger, sequentially-corrected alternative to test whether boosting could beat bagging.

All three trained with 5-fold stratified cross-validation and class-balanced weighting, then tuned via randomized search.

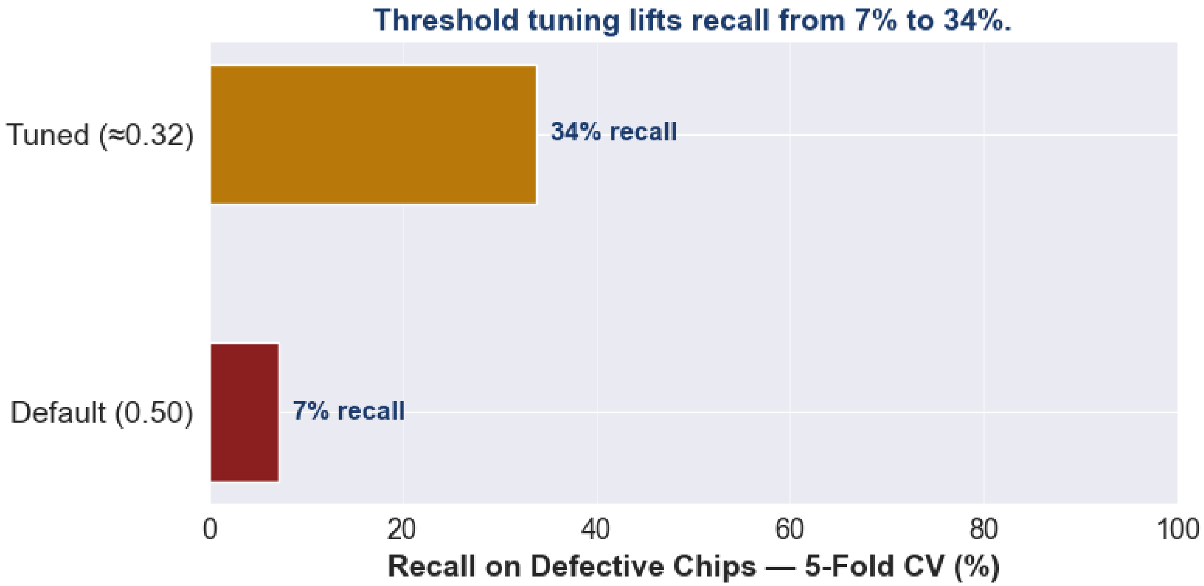
Random Forest was selected because it achieved the highest PR-AUC

Model	Recall	Precision	F1	ROC-AUC	PR-AUC
Logistic Regression	0.518	0.137	0.217	0.700	0.198
Random Forest	**0.071**	**0.467**	**0.122**	**0.769**	**0.285**
Gradient Boosting	0.085	0.463	0.141	0.725	0.246

PR-AUC measures how well a model ranks the rare Fail cases specifically, the metric that matters most here , and Random Forest led on it after tuning.



Threshold tuning improved detection of defective chips



Final Holdout Results (314 chips: 293 Pass / 21 Fail)

278 Correctly Passed	**15** False Alarms
16 Missed Defects	**5** Defects Caught

25%	24%	0.769	0.245
Precision (Fail)	Recall (Fail)	ROC-AUC	PR-AUC

314 holdout chips (293 Pass / 21 actual Fail), evaluated once. Recall (Fail) above is the holdout result, the chart's 34% is the cross-validated estimate used to choose the threshold.

How the sensors were prioritized

590 raw sensors

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Cleaning (missing data + near-zero-variance filtering)

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Statistical screening — narrowed to 75

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Model-based importance ranking

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Independent statistical cross-check

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Top 10 shortlist

The complete top-10 ranking, with statistical backup

Rank	Sensor	Importance	Std Dev	P-value
1	**feature_59**	**0.055**	**±0.014**	**0.000001**
2	**feature_341**	**0.029**	**±0.015**	**0.000209**
3	**feature_298**	**0.018**	**±0.007**	**0.017352**
4	feature_205	0.018	±0.011	0.000032
5	feature_587	0.017	±0.008	0.028323
6	feature_346	0.016	±0.002	0.041511
7	feature_40	0.016	±0.005	0.020053
8	feature_160	0.016	±0.004	0.020710
9	feature_431	0.016	±0.003	0.007320
10	feature_100	0.015	±0.004	0.005552

Importance = drop in held-out PR-AUC when the sensor is shuffled.
P-value = Mann-Whitney U test on training data only.